Training data generation
I_1
Iterative construction of 3D fractals following a random path
Starting at random pixel x_0
$x_{t+1} = A * x_t + \delta$
pixel position at $t+1$
random contraction matrix with bounded singular values
random translation vector
XY
YZ
XZ
Rigid field
random rotation
+
random translation
+
Gaussian random field
$\mathbf{h}$
Forward warp
Initial image $(x, I_1(x))$
apply displacement
displaced points $(x + \mathbf{h}(x), I_1(x))$
nearest neighbour interpolation
New image $(y, I_2(y))$
I_2
Model - anisotropic U-Net
Input stacked images
Output Displacement field
2
16
32
32
64
64
64
64
32
32
16
3
H, W D
H, W D
H/2, W/2 D
H/4, W,/4 D
H/8, W/8 D/2
H/16, W/16 D/4
H/16, W/16 D/4
H/8, W/8 D/2
H/4, W/4 D
H/2, W/2 D
H, W D
H, W D
2xconv 3x3x1
2x(conv 3x3x1, BatchNorm, GELU), max pool 2x2x1
2x(conv 3x3x3, BatchNorm, GELU), max pool 2x2x2
2x(conv 1x1x1, BatchNorm, GELU)
conv 1x1x1
2x(conv 3x3x3, BatchNorm, GELU), upsample 2x2x2
2x(conv 3x3x1, BatchNorm, GELU), upsample 2x2x1
Skip connection
Features
Height/Width multiplier
Depth multiplier
Features
64
48
32
16
Shape multiplier
1
1/2
1/4
1/8
1/16
Enc. 1
Enc. 2
Enc. 3
Enc. 4
Bottleneck
Dec. 1
Dec. 2
Dec. 3
Dec. 4
Network stage
Test data generation
I_1
Initial image
Mesh for Finite Element Method
XY
YZ
XZ
FEM: Physical model of linear elasticity
+
Rigid field
random rotation
+
random translation
$\mathbf{h}$
I_2
Deformed mesh + interpolation
Elastic deformation
Foward warp
Rigid deformation
Gaussian noise
Closer to real image

$$\mathcal{L}(\mathbf{h}) = \underbrace{\omega \|\mathbf{h}_{\mathrm{GT,mask}} - \mathbf{h}_{\mathrm{mask}}\|_2^2}_{\text{Supervision}} + \underbrace{\lambda \| I_1(x) - I_2(x + \mathbf{h})\|_2^2}_{\text{Warping quality}} + \underbrace{\alpha \|\nabla \mathbf{h}\|_{2,1}^2 + \beta \|\mathcal{H}\mathbf{h}\|_{2,1}^2}_{\text{Smoothness}}$$